

Data Analysis Capstone Project Expectations

For Data Preprocessing, EDA, Visualization and Dashboarding Module

Purpose: Learners should demonstrate the complete industry workflow of a Data Analyst: understanding a business problem, cleaning messy data, performing meaningful EDA, building an executive dashboard, and presenting actionable recommendations.

1. Business Understanding

Weightage: 10%

- Understand the business context and stakeholder need.
- Define clear project objectives and scope.
- Identify key business questions that the analysis must answer.
- Translate the problem into measurable data analysis outcomes.

Expected deliverable: Problem statement document of 1 to 2 pages.

2. Data Understanding

Weightage: 10%

- Understand all datasets, columns, data types and relationships.
- Prepare a data dictionary for important fields.
- Identify target KPIs and useful analytical dimensions.
- Explain what each dataset contributes to the business problem.

Expected deliverable: Data dictionary and dataset understanding summary.

3. Data Quality Assessment

Weightage: 15%

- Identify missing values, duplicates, invalid records and inconsistent formats.
- Detect outliers and unusual business values.
- Check incorrect data types, date formats and categorical inconsistencies.
- Document each issue along with the chosen resolution approach.

Expected deliverable: Data quality report with issue count, impact and resolution.

4. Data Cleaning and Preprocessing

Weightage: 15%

- Handle missing values using appropriate methods.
- Remove duplicates and standardize inconsistent categories.
- Treat outliers using justified business or statistical logic.

- Perform date-time processing, encoding, scaling or transformation where relevant.
- Create useful derived features and KPIs for analysis.

Expected deliverable: Clean dataset and Python or Jupyter notebook.

5. Exploratory Data Analysis

Weightage: 20%

- Perform univariate, bivariate and multivariate analysis.
- Use statistical summaries such as mean, median, standard deviation, percentiles and correlation.
- Create at least 15 meaningful visualizations.
- Write a clear insight for every important visualization.
- Connect analysis with the original business questions.

Expected deliverable: EDA notebook and EDA report with charts and insights.

6. Business Insights

Weightage: 10%

- Move beyond chart description and explain the business meaning.
- Highlight patterns, risks, bottlenecks, opportunities and exceptions.
- Use evidence from data to support each insight.
- Prioritize insights based on business impact.

Expected deliverable: Business insights report.

7. Dashboard Development

Weightage: 15%

- Build an executive-friendly dashboard using Power BI, Tableau or Streamlit.
- Include KPI cards, trends, comparisons and drill-down views.
- Add interactive filters such as date, region, department, product or category.
- Ensure the dashboard is clean, readable and decision-oriented.

Expected deliverable: Interactive dashboard file or published dashboard link.

8. Recommendations

Weightage: 5%

- Provide actionable and realistic recommendations.
- Map each recommendation to a data-backed observation.
- Mention expected business benefit where possible.
- Avoid generic suggestions that are not supported by analysis.

Expected deliverable: Recommendation summary table.